

As the Internet of Things becomes more of a reality, Contiki, an open source OS, allows DIY-enthusiasts to experiment with connecting tiny, low-cost, low-power microcontrollers to the Internet

Contiki is an open source operating system for connecting tiny, low-cost, low-power microcontrollers to the Internet. It is preferred because it supports various Internet standards, rapid development, a selection of hardware, has an active community to help and has commercial support bundled with an open source licence.

Contiki is designed for tiny devices and thus the memory footprint is far less when compared with other systems. It supports full TCP with IPv6, and the device's power management is handled by the OS. All the modules of Contiki are loaded and unloaded during run time; it implements protothreads, uses a lightweight file system and various hardware platforms with 'sleepy' routers (routers that sleep between message relays).

Contiki uses the Cooja simulator for emulation in case any of the hardware devices are not available.

Contiki can be downloaded as Instant Contiki, which is available in a single download that contains an entire Contiki development environment. It is a Ubuntu Linux virtual machine that runs in VMware player, and has Contiki and all the development tools, compilers and simulators used in Contiki development already installed on it. Most users prefer Instant Contiki over the source-code binaries. The current version of Contiki is 2.7.

The steps for installing Contiki OS are:

Step 1. Install VMware Player, which is free for academic and personal use.

Step 2. Download the Instant Contiki virtual image of size 2.5GB (approximately) from <http://sourceforge.net/projects/contiki/files/Instant%20Contiki/> and unzip it.

Step 3. Open the Virtual Machine and Contiki OS, and



Fig. 1: Contiki OS desktop

wait for the login screen to appear.

Step 4. Input the password as 'user' [this shows the desktop of Ubuntu (Contiki)].

To run a simulation, Contiki comes with many prebuilt modules that readily run on the Cooja simulator or on the real-hardware platform. There are two methods of opening the Cooja simulator window.

Method 1. On the desktop, as shown in Fig. 1, double-click on the Cooja icon. This will compile the binaries for the first time and open the simulation windows.

Method 2. Open the terminal and go to the Cooja directory:

```
pradeep@localhost$] cd contiki/tools/cooja
pradeep@localhost$] ant run
```

The simulation window is as shown in Fig. 2.

To create a simulation in Contiki, go to File menu→New Simulation, and name it as shown in Fig. 3.

Select any one radio medium (in this case)→Unit Disk Graph Medium (UDGM): Distance Loss and click Create.

Fig. 4 shows the simulation window, which has the following windows.

Network window. This shows all the notes in the simulated network.

Timeline window. This shows all the events over the time.

Mote output window. All serial port outputs are shown here.

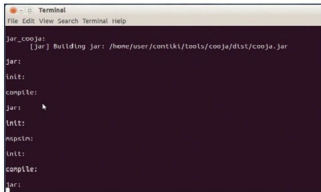


Fig. 2: Cooja compilation